Triangular Classifiers, k-Nearest Neighbors, and the Dependence on Density, Outliers, and Shape

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Abstract

One of the most common ways to classify a new point p in a labeled point set is k-NN (Nearest Neighbors). We analyze a situation where the labels are based on triangular separation in the plane and study how well k-NN performs in experiments using synthetic data. In particular, we analyze the dependence of the number of misclassifications on (i) the density of the labeled point set, (ii) the number of outliers of the labeled point set, and (iii) the perimeter of the triangular separation. The experiments show that (i) an increase in density in labeled points leads to a decrease in misclassifications, (ii) an increase in outlier fraction means an increase in misclassifications at above linear rate with f, and (iii) an increase in separator triangle scale leads to an increase in misclassifications.

1 Introduction

Classification is one of the most basic and important data mining tasks. Suppose we have a set of objects, each of which has a d-dimensional feature vector and one label. Classification is the task of assigning a label to any extra object of which we know only the feature vector (but not its label). Labels usually come from a finite set of classes, like {harmless, dangerous}, or {Catholic, Protestant, Buddhist, Islamic}, or {car, bicycle, truck}. When we do not have an application in mind, we often use colors for the classes. The feature vector contains a sequence of values that are attributes of the object which may influence the label. We can view the objects as points in a space, where the length of the feature vector is the dimension of the space. Hence, a set of objects used to build a classifier can be viewed as a set S of colored points in some space. Let us assume that in this space, we use the Euclidean metric to measure distances.

One of the most popular ways to perform classification of an object with known feature vector but unknown label is known as *k-Nearest Neighbor (k-NN) classification*. We view the object as a point p in space, and determine the closest k points from S. From these k points, we decide which color is used most often, and we assign that color to the point p. The main advantage of k-NN classification is its generality: it can be used regardless of the distribution of the colored points in S, and does not build a model for classification explicitly. There are many other methods for classification, some of which build a model explicitly. Each method has its pros and cons; for in-depth information we refer to standard texts on the topic [1, 2, 3, 4].

k-NN classification can also be used when a good model exists, but we do not know it. Suppose that we have a set of points in 2-dimensional space (the plane) with two colors, red and blue. Assume that in the plane, a triangle exists that perfectly separates the red and blue points: all points inside are blue and all points outside are red. If we use k-NN in this situation, our classifier will make some mistakes, or misclassifications. In particular, we may classify a point that lies inside the triangle but observe that more than half of the k nearest neighbors from S are red, or we may classify a point outside the triangle and find that the majority of the k nearest neighbors from S are blue. In this paper we study experimentally how well k-NN performs in different settings where a triangular separator exists. The different settings come from different densities of

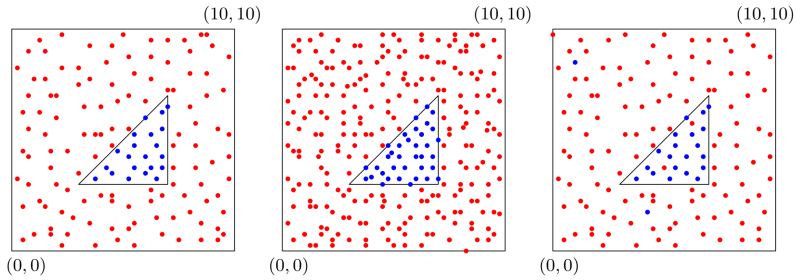

Figure 1: Examples of input sets to be generated for k-NN classification when a triangular separator exists. Left, a set S with no outliers. Middle, another set with double the density of S. Right, a set with the same density as S but with a few outliers.

the point set S, different numbers of outliers with respect to the triangular separator, and different shapes of the triangle.

We phrase the research questions precisely:

1. How does the number of misclassifications of k-NN depend on the density of red and blue points when a triangular separator for the red and blue points exists?
2. How does the number of misclassifications of k-NN depend on the fraction of outliers, when a triangular separator exists that has this fraction of outliers?
3. How does the number of misclassifications of k-NN depend on the perimeter of the triangular separator?

We answer these questions using synthetic data. We define a simple process that constructs sets S as desired.

The remainder of this paper is organized as follows. In Section 2 we describe the experimental set-up. In Section 3 we show the outcome of the experiments, make observations, and evaluate the results. In Section 4 we conclude the paper and suggest possibilities for further research.

2 Experimental set-up

We use the following set-up for our experiments. Let Q be the square $[0, 10] \times [0, 10]$, and let T be the triangle with vertices $(3, 3)$, $(7, 3)$ and $(7, 7)$. We denote the number of points in S by n and we denote the outlier fraction by f. We generate S by creating n points uniformly distributed, at random, inside Q. If a generated point lies inside T it is assigned the color blue, and otherwise it is assigned the color red. However, with probability f, we reverse the color of the generated point by making it red when it is inside T and blue when it is outside.

As default settings we set $n = 800$, $k = 7$, and $f = 0$. To influence the density we vary n, and to influence the fraction of outliers we vary f.

A single experiment consists of generating one set S with certain parameters, and then generating 10,000 points uniformly at random inside Q as test points. For each point $p \in Q$, we determine its k nearest neighbors, and thereby the color obtained from k-NN classification. We also determine whether p lies inside T. If p lies inside T but is classified as red, we say that p is misclassified. If p lies outside T but is classified as blue, it is also misclassified. We compute the number of misclassified points among the $10,000$ randomly generated ones in Q. Each experiment is repeated 20 times to get 20 values for an experiment using the same settings, but different coordinates of points in S, different coordinates of the 10,000 test points, and, if $f > 0$, different outliers.

To answer the first research question—the dependence of the number of misclassified points on n—we let n take on the values 100, 200, 300, 400, 500, 600, 700, 800, 900, 1000. The other parameters k and f use their default value.

To answer the second research question—the dependence of the number of misclassified points on f—we let f take on the values 0.0, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3. The other parameters k and n use their default value.

To answer the third research question—the dependence of the number of misclassified point on the perimeter of the triangular separator—we adjust the scale of the triangle by multiplying it with the values 0.5, 0.75, 1 1.25, 1.5, 1.75 and 2. These configurations are still within the boundaries of Q. The other parameters k, f and n use their default values.

3 Results and analysis

In this section, we show the results of the experiments and discuss what they imply for the answers to the research questions. We treat the three research questions consecutively.

3.1 Experiment 1: dependence on density

Experiment 1 is designed to explore how the number of misclassifications of k-NN depends on the density of labeled points when a triangular separator for the labeled points exists. We varied the density of the labeled point set by increasing the number of labeled points from 100 to 1000 by 100 in ten experiments repeated 20 times each. The other parameters k and f use their default values. After running the code to simulate Experiment 1, we get the mean and the standard deviation of the number of misclassified points for each experiment, which are shown in Table 3.1. The visualization of the data is shown in Figure 2.

The results indicate that the average number of misclassifications significantly decreases as the number of labeled points increases from 100 to 500, and then the misclassification count declines steadily. This may be caused by more labeled point density means that the classification boundaries can be represented more accurately, thus improving the performance of the k-NN classification on data sets that admit a triangular classifier. The standard deviation also decreases as the number of labeled points grows, indicating that the variability in misclassification counts diminishes with more labeled points. This implies that higher-density labeled data sets yield more stable misclassification results across experiments. In summary, k-NN performs better on data sets that admit a triangular classifier as the density of labeled point set increases.

number of labeled points	Mean misclassifications	Standard deviation
100	480.45	127.61
200	334.1	69.47
300	283.1	67.36
400	232.65	46.46
500	206.55	31.62
600	199.65	30.49
700	172.9	21.89
800	168.05	29.12
900	158.95	36.76
1000	153.5	28.43

Table 1: Exact values of Mean and Standard Deviation found across 20 runs for each n-value.

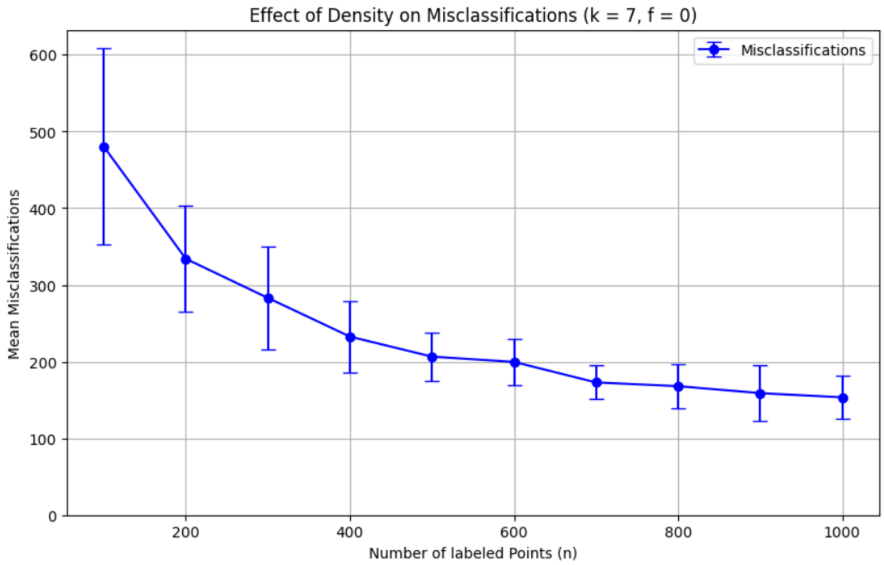

Figure 2: Plot of the Mean and Standard Deviation found across 20 runs for each n-value.

3.2 Experiment 2: dependence on outlier fraction

For experiment 2 we ran the tests using the default parameters $n = 800$, $k = 7$. For the value of f, we ran the test on each of the values in [0.00, 0.05, 0.10, 0.15, 0.20, 0.25, 0.30]. For each value, we ran the test 20 times using random generation of point locations and took the mean misclassifications as well as the standard deviation of the misclassifications to the mean misclassification. See figure 3 and table 2 for the data obtained.

We observe that the amount of misclassifications increases non-linearly with the outlier fraction value f. This follows intuition, as the number of outliers increases, each point p in the randomly generated 10.000 points is more likely to have mislabeled neighbours from n. The higher the amount of misclassified neighbours, the more likely the point p is to be misclassified using k-NN. This effect is quite minor at lower f-values as k is set to 7, requiring 4 mislabeled neighbours to be misclassified, the likelihood of which is quite low when f is low. But as the f-value increases points the likelihood of each point p in the generated points to have 4 or more misclassified neighbours increases non-linearly with f causing non-linear growth of misclassifications with f.

Another observation is that the standard deviation increases with each step of increasing f. When looking at the ratio between standard deviation relative to the mean misclassification count, we find that the ratio stays relatively similar within a range of 0.15 to 0.20. This means that the deviation likely increases linearly with the amount of misclassifications. This seems like quite natural behaviour since as f increases, the variance of observed misclassification rates stay the same, but absolute values observed increase.

Though no further testing was done on other k-values due to time constraints, we expect the value of k to have an effect on the growth-rate of misclassifications with f. In cases where k is low, it is easier for points p to be misclassified at lower f-values. We expect that lower k values would result in more linear growth and higher k-values to cause more non-linear where turning points towards large growth in misclassification rates appear at higher f-values as k-values increase.

f-value	Mean misclassifications (M)	Standard deviation (σ)	σ over M
0.00	167.34	25.17	0.15
0.05	187.26	31.47	0.17
0.10	231.06	43.10	0.19
0.15	344.85	71.85	0.21
0.20	549.16	111.51	0.20
0.25	930.54	185.46	0.20
0.30	1447.78	243.36	0.16

Table 2: Values of mean misclassifications (M) and standard deviation (σ) as well as σ over M found across 20 runs for each f-value

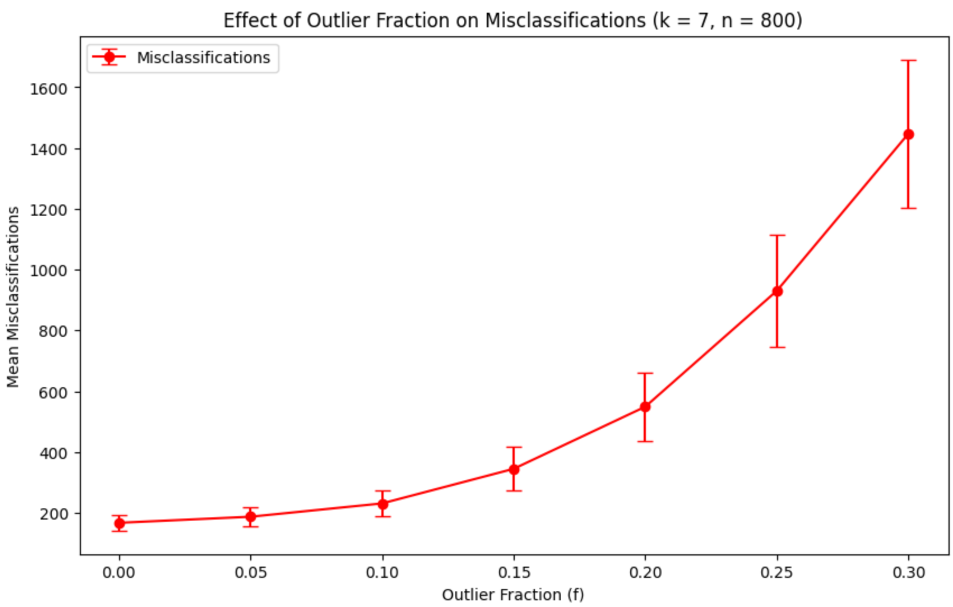

Figure 3: Plot of mean misclassifications and standard deviation found across 20 runs for each f-value

3.3 Experiment 3: dependence on triangle perimeter

The last experiment will indicate whether a change in the triangle perimeter will have an effect on the amount of misclassifications within the model. The change in perimeter we will look at surrounds the scale of the triangle. We will look at 7 different sizes of the original triangle, where the original size is included. The scale values are 0.5, 0.75, 1 (original), 1.25, 1.5, 1.75 and 2, which are all within the boundaries of Q. The other parameters k, f, and n use their default values. As with the other experiments, for each of the scale values we will perform 20 tests and take the mean and standard deviation based on those tests. A representation of the results is shown in Figure 4, with the corresponding exact values of the means and standard deviations in Table 3.

Based on the visualization of the means and standard deviations, there is a clear trend in the effect of scale on the perimeter of the triangular separator. The mean of misclassifications grows steadily with each scale increase. The standard deviation doubles after the original scale of 1. However, at scale 1.75 de standard deviation goes back down. This inconsistency does not allow us to determine whether the standard deviations increase with the scale as well. This could also be an outlier considering the other scales do increase. The increase of the mean is likely due to the increased range of the edges of the triangle. Since the edges become longer there are more opportunities for testing points to become misclassified. Therefore, we can conclude that an increase in the scale of the perimeter increases the number of misclassifications by the k-NN model.

Scale	Mean misclassifications	Standard deviation
0.5	80.8	20.62
0.75	119.9	18.48
1	171.85	20.05
1.25	213.2	34.579
1.5	266.2	35.34
1.75	281.25	22.71
2	339.75	39.879

Table 3: Exact values of Mean and Standard Deviation found across 20 runs per scale change

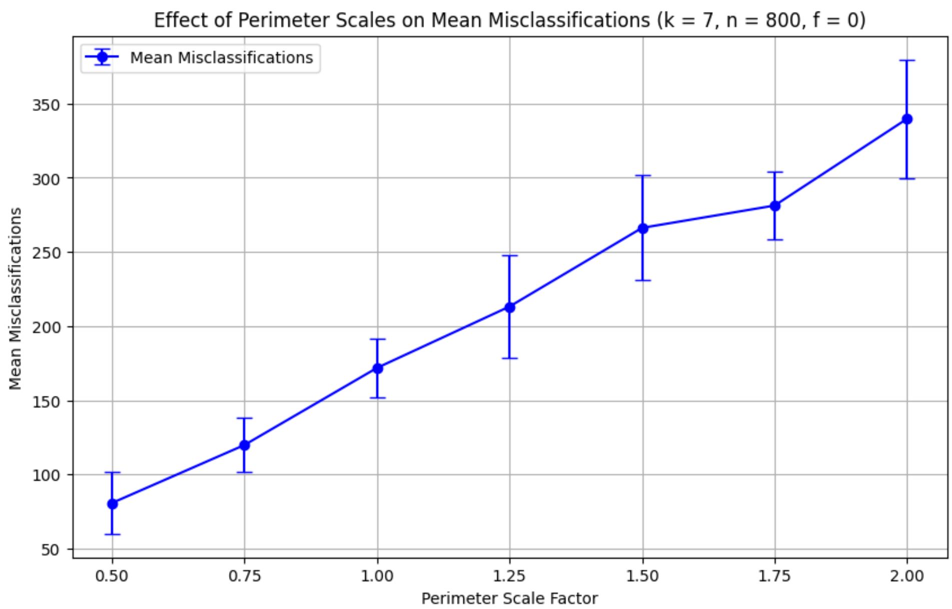

Figure 4: Plot of the Mean and Standard Deviation found across 20 runs per scale change

4 Conclusions and further research

To summarize, we looked at the dependency of misclassifications of k-NN on different parameters, namely the density of labeled points, the fraction of outliers, and the perimeter of the triangular separator. In the first experiment, we measured the number of misclassifications for each experiment with the different number of labeled points. After visualization, we observed that an increase in density of n meant a decrease in misclassifications. In the second experiment, we measured misclassification rates of generated points p at different f-values and found that the number of misclassifications of k-NN depends non-linearly on the fraction of outliers, increasing at above linear rates as the fraction of outliers increases for $k = 7$. In the last experiment, we concluded that an increase in the scale of the perimeter increased the amount of misclassifications of the k-NN model.

For further research we suggest looking into how different values of k influence the number of misclassifications across each parameter, to get a better view of how k-NN classification accuracy depends on said parameters.

Another suggestion would be to see how the k-NN classification performs across other separator shapes such as circles, rectangles or other polygons to research if k-NN classification works better on separators that have particular geometric properties.

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